ECE 3210 hw01

1. Consider the following signals defined using unit step functions.
   (a) Sketch the following function
   $$f_1(t) = t^2\big[u(t-1) - u(t-2)\big].$$
   (b) Sketch the following function
   $$f_2(t) = (t-4)\big[u(t-2) - u(t-4)\big].$$

2. Evaluate the following integrals. Some answers will be a numerical value, while others will be a function of t.

   (a)
   $$\int_{-\infty}^{\infty} \delta(t)\, e^{-j\omega t}\, dt$$

   (b)
   $$\int_{-\infty}^{\infty} \delta(2t-3)\, \sin(\pi t)\, dt$$

   (c)
   $$\int_{-\infty}^{\infty} \delta(t+3)\, e^{-t}\, dt$$

   (d)
   $$\int_{-\infty}^{\infty} (t^3+4)\, \delta(1-t)\, dt$$

   (e)
   $$\int_{-\infty}^{\infty} \delta(\tau)\, \sin(t-\tau)\, d\tau$$

   (f)
   $$\int_{-\infty}^{\infty} \sin(\tau)\, \delta(t-\tau)\, d\tau$$

3. Consider the signal
$$x(t) = \begin{cases} t & 0 \le t < 1 \\ 0.5 + 0.5\cos(2\pi t) & 1 \le t < 2 \\ 3 - t & 2 \le t < 3 \\ 0 & \text{otherwise} \end{cases}$$
The energy of $x(t)$ is approximately $E_x \approx 1.0417$.
   (a) What is the energy of $y_1(t) = \frac{1}{3}\,x(2t)$?
   (b) A periodic signal $y_2(t)$ is defined with period $T = 4$ s as
$$y_2(t) = \begin{cases} x(t) & 0 \le t < 4 \\ y_2(t+4) & \forall\, t \end{cases}.$$
   What is the *power* of $y_2(t)$?
   (c) What is the *power* of $y_3(t) = \frac{1}{3}\,y_2(2t)$?

4. For each system below with input $x(t)$ and output $y(t)$, determine whether the system is **linear** or **nonlinear**.

   (a)
   $$\frac{dy(t)}{dt} + 2y(t) = x^2(t)$$

   (b)
   $$\frac{dy(t)}{dt} + 3t\,y(t) = t^2 x(t)$$

   (c)
   $$5y(t) + 1 = x(t)$$

   (d)
   $$\left(\frac{dy(t)}{dt}\right)^2 + 2y(t) = x(t)$$

   (e)
   $$\frac{dy(t)}{dt} + \sin(t)\,y(t) = \frac{dx(t)}{dt} + 2x(t)$$

   (f)
   $$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} + 2x(t)$$

5. For each system below with input $x(t)$ and output $y(t)$, determine whether the system is **time-invariant (TI)** or **not time-invariant (not TI)**.

   (a)
   $$y(t) = x(t-2)$$

   (b)
   $$y(t) = \log\big(x(t)\big)$$

   (c)
   $$y(t) = x(t)^2$$

   (d)
   $$y(t) = \int_{-5}^{5} x(\tau)\, d\tau$$

   (e)
   $$y(t) = t\, x(t)$$

   (f)
   $$y(t) = \cos(t)\, x(t) + a$$

6. A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^3 + 9D)\, y(t) = (2D^3 + 1)\, x(t).$$

   (a) Find the characteristic equation $Q(\lambda) = 0$ of the system.

   (b) Given the initial conditions $y_{\text{zir}}(0) = 4$, $y'_{\text{zir}}(0) = -18$, and $y''_{\text{zir}}(0) = 0$, determine the zero-input response $y_{\text{zir}}(t)$ for $t \geq 0$.

7. A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^2 + 4D + 4)\, y(t) = D\, x(t).$$

   (a) Find the characteristic equation $Q(\lambda) = 0$ of the system.

   (b) Find the characteristic roots (eigenvalues).

   (c) Given the initial conditions $y_{\text{zir}}(0^-) = 3$ and $y'_{\text{zir}}(0^-) = -4$, determine the zero-input response $y_{\text{zir}}(t)$ for $t \geq 0$.

Answer Key

1(a) $f_1(t) = t^2$ for $1 \leq t < 2$, zero otherwise

1(b) $f_2(t) = t - 4$ for $2 \leq t \leq 4$, zero otherwise

2(a) 1

2(b) $-\frac{1}{2}$

2(c) e^3

2(d) 5

2(e) $\sin(t)$

2(f) $\sin(t)$

3(a) 0.057872

3(b) $P_{y_2} \approx 0.2604$

3(c) $P_{y_3} \approx 0.0289$

4(a) Nonlinear

4(b) Linear

4(c) Nonlinear

4(d) Nonlinear

4(e) Linear

4(f) Linear

5(a) Time-invariant

5(b) Time-invariant

5(c) Time-invariant

5(d) Not time-invariant

5(e) Not time-invariant

5(f) Not time-invariant

6(a) $Q(\lambda) = \lambda^3 + 9\lambda$

6(b) $y_{\text{zir}}(t) = 4 - 6\sin(3t), \quad t \geq 0.$

7(a) $Q(\lambda) = \lambda^2 + 4\lambda + 4$

7(b) $\lambda_{1,2} = -2$ (repeated)

7(c) $y_{\text{zir}}(t) = (3 + 2t)\, e^{-2t}, \quad t \geq 0$